

NAME : SHAH FAISAL

ROLL NO : 23I-0058

SECTION : BSE – 6A

Computer Networks LAB 05

BSSE SPRING 2026

LAB 05

TASK 01

File C:/Users/Shah%20Faisal/Downloads/i230058_Shah_Faisal.html

Shah Faisal - i230058

Lab 5: Secure Digital Transmissions

Test Login Form (HTTP)

Username (Enter your Roll Number):

Password:

Age:

Submit (HTTP POST)

Instructions for Wireshark Capture:

1. Open Wireshark and start capturing on your network interface
2. Use filter: host httpbin.org and http
3. Fill out this form with test credentials
4. Click submit
5. Look for the HTTP POST packet in Wireshark

Ln 4, Col 7 100% Windows (CRLF) UTF-8

The Wireshark Network Analyzer

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Welcome to Wireshark

Capture

...using this filter: All interfaces shown

- WiFi
- VMware Network Adapter VMnet8
- VMware Network Adapter VMnet1
- Ethernet
- Adapter for loopback traffic capture
- Local Area Connection* 8**
- Local Area Connection* 7
- Local Area Connection* 6
- Local Area Connection* 9
- Event Tracing for Windows (ETW) reader: etwdump

Learn

[User's Guide](#) · [Wiki](#) · [Questions and Answers](#) · [Mailing Lists](#) · [SharkFest](#) · [Wireshark Discord](#) · [Donate](#)

You are running Wireshark 4.6.3 (v4.6.3-0-g648f69f3e168). You receive automatic updates.

Ready to load or capture || No Packets ||

Ln 4, Col 7 100% Windows (CRLF) UTF-8

The screenshot displays two windows. The top window is Wireshark, titled 'Capturing from WiFi'. It shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The selected packet is No. 1289, a TCP Reset (RST) from Apple_68:ee:d5 to Apple_68:ee:d5. The packet details pane shows the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol (RST) layers. The packet bytes pane shows the raw data in hexadecimal and ASCII.

The bottom window is Notepad, titled 'Untitled - Notepad'. It contains the following text:

```
Name Shah Faisal
Roll Number 23I-0058
SECTION ; BSE-6A
lab 05
```

Wireshark - Packet 1789 - WiFi

> Frame 1789: Packet, 633 bytes on wire (5064 bits), 633 bytes captured (5064 bits) on interface \Device\NPF_{81...}

> Ethernet II, Src: Apple_68:ee:d5 (88:e9:fe:68:ee:d5), Dst: HewlettPacka_d0:48:00 (08:97:34:d0:48:00)

> Internet Protocol Version 4, Src: 172.15.73.218, Dst: 44.195.71.76

> Transmission Control Protocol, Src Port: 51939, Dst Port: 80, Seq: 1, Ack: 1, Len: 579

> Hypertext Transfer Protocol

▼ HTML Form URL Encoded: application/x-www-form-urlencoded

> Form item: "username" = "1230058"

> Form item: "password" = "123456"

> Form item: "age" = "22"

0000 08 97 34 d0 48 00 88 e9 fe 68 ee d5 08 00 45 00 ..4.H...h...E
 0010 02 6b 4f e1 40 00 80 06 3e b3 ac 0f 49 da 2c c3 .kO@...-...I.
 0020 47 4c ca e3 00 50 c3 27 0d ac 40 47 f1 e1 50 18 GL...P.'@G.P

No.: 1789 - Time: 95.186765 - Source: 172.15.73.218 - Destination: 44.195.71.76 - Protocol: HTTP - Length: 633 - Info: POST /post HTTP/1.1 (application/x-www-form-urlencoded)

☒ Show packet bytes Layout: Vertical (Stacked)

Close Help

WiFi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No. Time Source Destination Protocol Length Info

References Mailings Review View Help Easy Syntax Highl

File Edit Format View Help

Name Shah Faisal
Roll Number 23I-0058
SECTION ; BSE-6A
lab 05

*Untitled - Notepad

***WiFi**

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
186765	172.15.73.218	44.195.71.76	HTTP	633	POST /post HTTP/1.1 (application/x-www-form-	
405062	44.195.71.76	172.15.73.218	HTTP/J...	1178	HTTP/1.1 200 OK , JSON (application/json)	
1821	95.481344	172.15.73.218	HTTP	424	GET /favicon.ico HTTP/1.1	
1823	95.697983	44.195.71.76	HTTP	287	HTTP/1.1 404 NOT FOUND (text/html)	

Frame 1789: Packet, 633 bytes on wire (5064 bits), 633 bytes

Ethernet II, Src: Apple_68:ee:d5 (88:e9:fe:68:ee:d5), Dst: He

Internet Protocol Version 4, Src: 172.15.73.218, Dst: 44.195.

Transmission Control Protocol, Src Port: 51939, Dst Port: 80,

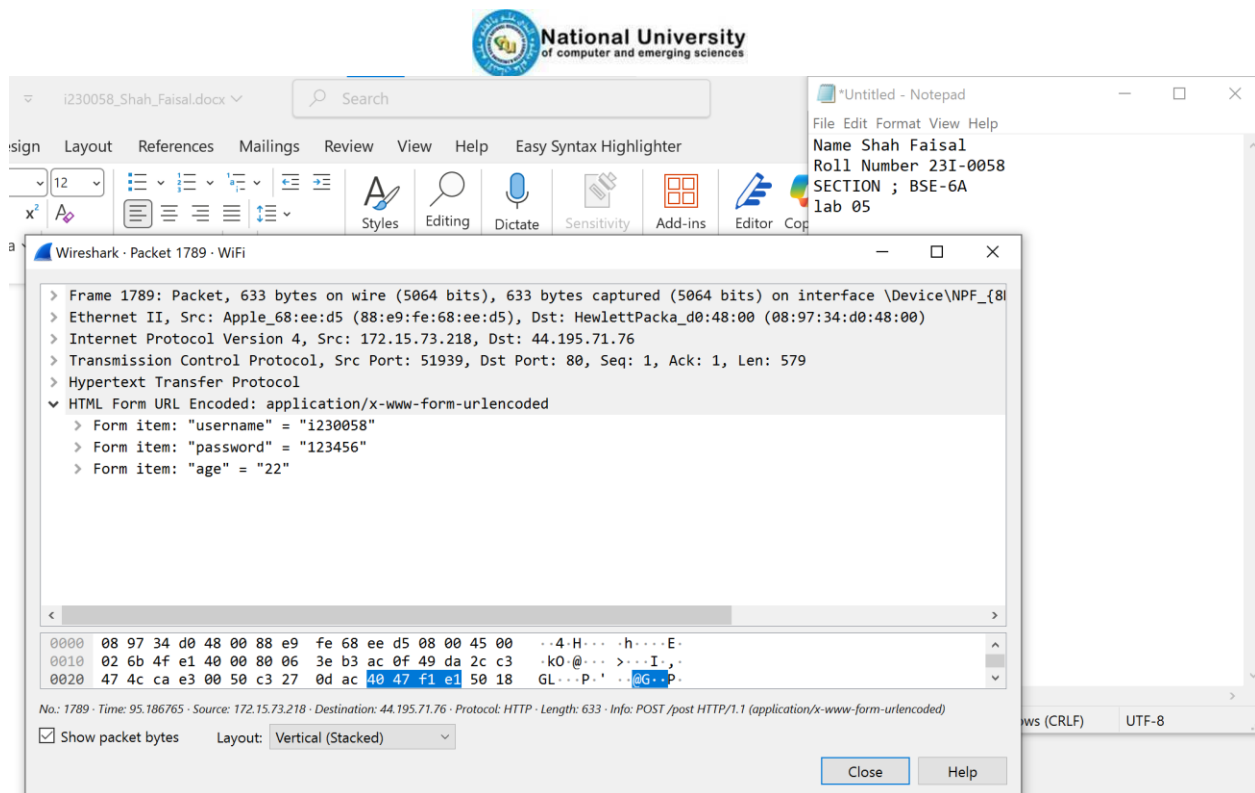
Hypertext Transfer Protocol

HTML Form URL Encoded: application/x-www-form-urlencoded

```

0000 08 97 34 d0 48 00 88 e9 fe 68 ee d5 08 00 45 00  ..4
0010 02 6b 4f e1 40 00 80 06 3e b3 ac 0f 49 da 2c c3  ..kC
0020 47 4c ca e3 00 50 c3 27 0d ac 40 47 f1 e1 50 18  GL.
0030 02 00 d6 ed 00 00 50 4f 53 54 20 2f 70 6f 73 74  ...
0040 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a  HT
0050 20 68 74 74 70 62 69 6e 2e 6f 72 67 0d 0a 43 6f  ht
0060 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61  nne
0070 6c 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65  liv
0080 6e 67 74 68 3a 20 33 39 0d 0a 43 61 63 68 65 2d  ngt
0090 43 6f 6e 74 72 6f 6c 3a 20 6d 61 78 2d 61 67 65  Cor
00a0 3d 30 0d 0a 4f 72 69 67 69 6e 3a 20 6e 75 6c 6c  =0.
00b0 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20  ..C
00c0 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 2d 77 77  app
00d0 77 2d 66 6f 72 6d 2d 75 72 6c 65 6e 63 6f 64 65  w-f

```



Q1. Why is HTTP considered insecure for transmitting sensitive data? Explain the risks of plaintext transmission and how attackers can exploit it.

HTTP transmits data as plaintext without encryption, meaning anyone with access to the network (using tools like Wireshark) can intercept and read sensitive information such as usernames, passwords, and personal details. Attackers on the same Wi-Fi network can easily capture this unencrypted data through packet sniffing, leading to credential theft, identity fraud, and other cybersecurity breaches.

Task 02

← → ↻ 🌐 wikipedia.org ☆ ⬇️ S School ⋮

WIKIPEDIA

25 years of the free encyclopedia

English
7,125,000+ articles

日本語
1,487,000+ 記事

Deutsch
3,090,000+ Artikel

Русский
2 081 000+ статей

Français
2 734 000+ articles

Español
2 088 000+ artículos

Italiano
1,953,000+ voci

中文
1,521,000+ 条目 / 條目

Português
1,163,000+ artigos

Polski
1 682 000+ haseł



Who is hiding in the globe?

A special new friend is waiting inside the puzzle globe. Uncover who it is and learn what other surprises are in store for Wikipedia's 25th birthday.

EN 🔍

https://fr.wikipedia.org

🌐 Read Wikipedia in your language

12:06 PM

*Untitled...

File Edit Format View Help

Name Shah Faisal
Roll Number 23I-0058
SECTION ; BSE-6A
lab 05

Windows (CRLF) UTF-8

The Wireshark Network Analyzer

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Welcome to Wireshark

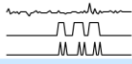
Capture

...using this filter: Enter a capture filter ... All interfaces shown

☐ Local Area Connection* 8

Local Area Connection* 7
Local Area Connection* 6
Local Area Connection* 9
VMware Network Adapter VMnet1
Local Area Connection* 10
Ethernet

☒ Event Tracing for Windows (ETW) reader: etwdump



Learn

[User's Guide](#) · [Wiki](#) · [Questions and Answers](#) · [Mailing Lists](#) · [SharkFest](#) · [Wireshark Discord](#) · [Donate](#)

You are running Wireshark 4.6.3 (v4.6.3-0-g648f69f3e168). You receive automatic updates.

Ready to load or capture | No Packets | Profile: Default

en.wikipedia.org/wiki/Tomato



School

Wiki Loves Folklore

Photograph your local culture, help Wikipedia and win!

PARTICIPATE NOW



Tomato

197 languages

Article Talk

Read View source View history Tools

From Wikipedia, the free encyclopedia

For other uses, see [Tomato \(disambiguation\)](#).

The **tomato** (US: /təˈmeɪtoʊ/, UK: /təˈmɑːtoʊ/; ***Solanum lycopersicum***) is a plant whose fruit is an edible berry that is eaten as a vegetable. The tomato is a member of the nightshade family that includes tobacco, potato, and chili peppers. It originated from western South America, and may have been domesticated there or in Mexico (Central America). It was introduced to the Old World by the Spanish in the Columbian exchange in the 16th century.

Tomato plants are vines, largely annual and vulnerable to frost, though sometimes living longer in greenhouses. The flowers are able to self-fertilise. Modern varieties have been bred to ripen uniformly red, in a process that has impaired the fruit's sweetness and flavor. There are thousands of cultivars, varying in size, color, shape, and flavor. Tomatoes are attacked by many

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

en.wikipedia.org/wiki/Tomato



School

Wiki Loves Folklore

Photograph your local culture, help Wikipedia and win!

PARTICIPATE NOW



Tomato

197 languages

Article Talk

Read View source View history Tools

From Wikipedia, the free encyclopedia

For other uses, see [Tomato \(disambiguation\)](#).

The **tomato** (US: /təˈmeɪtoʊ/, UK: /təˈmɑːtoʊ/; ***Solanum lycopersicum***) is a plant whose fruit is an edible berry that is eaten as a vegetable. The tomato is a member of the nightshade family that includes tobacco, potato, and chili peppers. It originated from western South America, and may have been domesticated there or in Mexico (Central America). It was introduced to the Old World by the Spanish in the Columbian exchange in the 16th century.

Tomato plants are vines, largely annual and vulnerable to frost, though sometimes living longer in greenhouses. The flowers are able to self-fertilise. Modern varieties have been bred to ripen uniformly red, in a process that has impaired the fruit's sweetness and flavor. There are thousands of cultivars, varying in size, color, shape, and flavor. Tomatoes are attacked by many

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes

Clade: Angiosperms

Clade: Eudicots

Appearance hide

Text

☐ Small

☒ Standard

☐ Large

Width

☒ Standard

☐ Wide

Color (beta)

☐ Automatic

☒ Light

☐ Dark

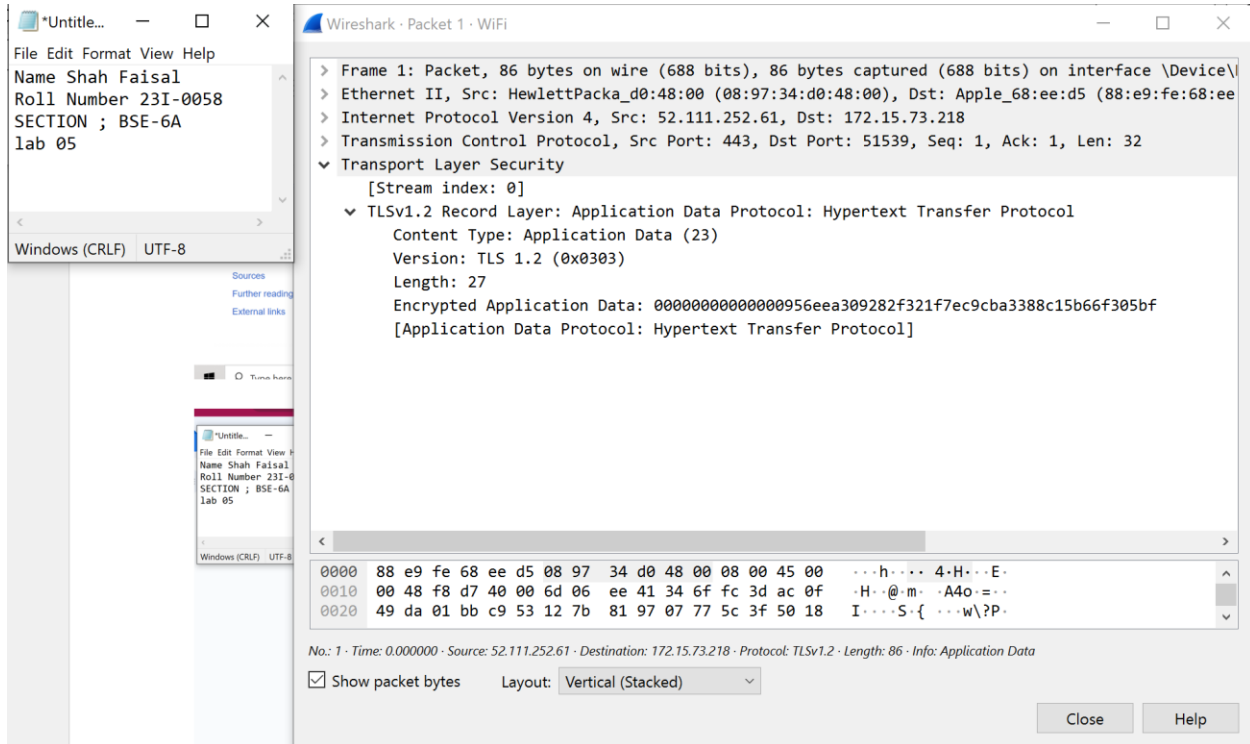
Tomato



Scientific classification

Kingdom: Plantae

Clade: Tracheophytes



Q1. What differences do you notice between HTTP and HTTPS traffic? Compare the packet details for HTTP vs. HTTPS and note how the data is secured.

HTTP traffic displays all form data (username, password, age) in plain readable text under the "HTML Form URL Encoded" section, while HTTPS traffic shows only "Encrypted Application Data" as unreadable hexadecimal values under TLS sections, confirming that HTTPS encrypts the actual content before transmission.

Q2. Why is HTTPS considered more secure than HTTP? Explain the role of SSL/TLS encryption in preventing data interception and tampering.

HTTPS uses SSL/TLS encryption to scramble data into unreadable format before transmission, ensuring that even if attackers intercept the packets, they only see encrypted garbage instead of actual credentials. This encryption also includes integrity checks that prevent tampering, as any modification to the encrypted data during transmission would be detected by the recipient.